Stochastic Modelling Project 2

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1 Problem 1: Urgent Care Center

A small urgent care center (UCC) has capacity to treat only one patient at a time. Assume that the arrival of patients to the UCC follows a Poisson process with rate $\lambda > 0$, that treatment times are independent and exponentially distributed with mean $1/\mu > 0$, and that treatment times are independent of the arrival process.

If no one is receiving treatment, an arriving patient immediately begins treatment. If another patient is currently being treated, the arriving patient joins the end of the queue. For the purpose of this project, we assume that there is no upper limit on the number of patients that can wait inside the UCC.

Assume that $\lambda < \mu$, and let $X(t)$ denote the number of patients in the UCC at time t. We also know that $X(0) = 0$.

1.1 Defining the stochastic process

The conditions for an M/M/1 queue are:

1. Interarrival times are independent and identically distributed as Exp(λ).
2. Service times are independent and identically distributed as Exp(μ).
3. There is one server, and service times are independent of the arrival process.

The arrival process is Poisson with rate λ. The interarrival time, i.e., the time until the next patient arrives, is exponentially distributed with rate parameter λ. Thus, condition (1) holds. Condition (2) is satisfied by assumption. Because both (1) and (2) are satisfied, and only one patient can be treated at a time, condition (3) also holds. Therefore, the process is an M/M/1 queue.

The process $\{X(t) : t \geq 0\}$ can be viewed as a birth–death process because:

- The possible states are $\{0, 1, 2, \ldots\}$.
- Transitions from state i can only occur to $i+1$ (a patient arrives) or $i-1$ (a patient completes treatment).

The transition (birth and death) rates are:

$$\lambda_i = \lambda, \quad \mu_i = \mu.$$

By Little's Law, the average time a patient spends at the UCC is:

$$W = \frac{L}{\lambda},$$

where L is the average number of patients in the system. Since $\mu > \lambda$, the queue is stable.

For an M/M/1 queue, the stationary distribution is geometric with parameter $1 - \frac{\lambda}{\mu}$. Therefore,

$$L = \mathrm{E}\left[\text{Geometric}\left(1 - \frac{\lambda}{\mu}\right)\right] = \frac{\lambda}{\mu - \lambda}, \tag{1}$$

and

$$W = \frac{1}{\mu - \lambda}. \tag{2}$$

1.2 Simulation

Assume now that $\lambda = 5$ patients per hour and that the expected treatment time is 10 minutes, i.e., $\frac{1}{\mu} = \frac{1}{6}$ hours, so $\mu = 6$. According to (1) and (2), the corresponding long-run mean number of patients and mean time spent at the UCC is $L = 5$ patients and $W = 1$ hour. We simulate $\{X(t) : t \geq 0\}$ over 50 days (1200 hours) 30 times to examine these quantities. The resulting 95% confidence intervals are:

$$L \in [4.905, 5.387] \text{ and } W \in [0.981, 1.077]$$

We find L via the mean number of patients at the UCC, and W through L via little's law. The 95% CIs are calculated through the t-distribution.

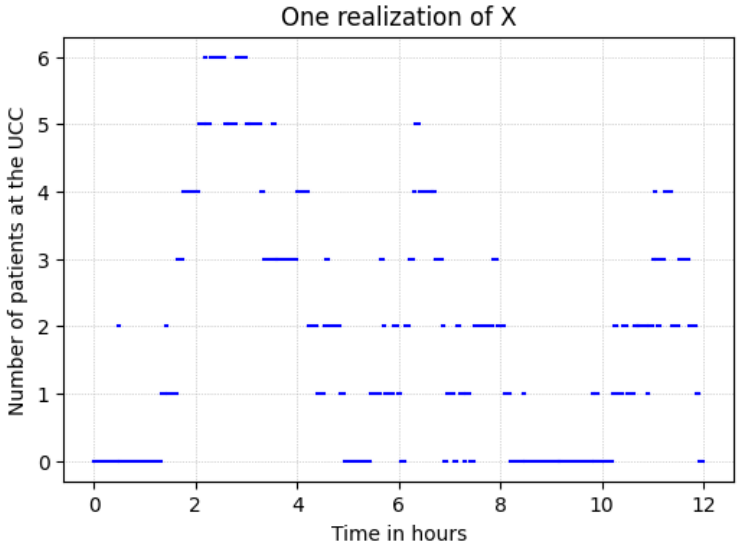


Figure 1: Temporal evolution of the number of patients in the UCC in the span of 12 hours

1.3 Urgent vs. Normal Patients

The model above is unrealistic, as a UCC typically operates with a triage system where patients are prioritized by urgency. Assume that upon arrival, each patient is independently classified as either urgent (with probability p, where $0 < p < 1$) or normal (with probability $1 - p$).

The queueing discipline is as follows:

- Urgent patients have priority over normal patients.
- If a normal patient is being treated when an urgent patient arrives, treatment is preempted: the urgent patient immediately begins treatment, and the interrupted normal patient returns to the normal queue.

Let $U(t)$ and $N(t)$ denote the number of urgent and normal patients in the system at time t.

Urgent patients

The urgent patients form an M/M/1 queue. Because urgent patients preempt treatment over normal patients, the normal patients are effectively invisible to the urgent patients. Thus, the urgent-process by itself is identical to the previous process. By the thinning property of Poisson processes, the arrival rate of urgent patients is $p\lambda$. The service rate remains μ. Thus, $\{U(t)\}$ satisfies the M/M/1 conditions.

The long-run mean number of urgent patients in the system is:

$$L_U = \frac{p\lambda}{\mu - p\lambda}. \tag{3}$$

Normal patients

The process $\{N(t)\}$ does not form an M/M/1 queue because it is not memoryless. For example, if we know that the system transitioned from state 2 to 1 (i.e., a normal patient was served), we can infer that there were no urgent patients waiting. In this case, this additional information (i.e., conditioning on past states) increases our knowledge of the system and consequently our ability to forecast future states. Hence, the Markov property does not hold for $\{N(t)\}$.

However, we may still find the long-run mean number of normal patients in the system. Since the total number of patients at all times is $X(t) = U(t) + N(t)$, the long-run mean number of normal patients can be obtained through (1) and (3):

$$L_N = L - L_U = \frac{\lambda}{\mu - \lambda} - \frac{p\lambda}{\mu - p\lambda} = \frac{(1-p)\lambda\mu}{(\mu - \lambda)(\mu - p\lambda)}. \tag{4}$$

1.4 Expected time in system

Applying Little's Law again, the average time spent at the UCC is:

For urgent patients:

$$W_U = \frac{L_U}{p\lambda} = \frac{1}{\mu - p\lambda}. \tag{5}$$

For normal patients:

$$W_N = \frac{L_N}{(1-p)\lambda} = \frac{\mu}{(\mu - \lambda)(\mu - p\lambda)}. \tag{6}$$

Time Spent at the UCC as a Function of p

As p increases, more patients are classified as urgent. This has implications on the total waiting time for patients in the system. When $p \approx 0$, there are mostly normal patients coming in, and we end up with effectively the same model that we first examined, where $W = 1/\mu = 1$ hour. The opposite case, where $p \approx 1$, the arrivals are effectively always urgent. Looking at urgent patients in isolation, the system is again similar to a regular M/M/1 queue. The few normal patients that arrive effectively never receive treatment, as there are almost always urgent patients who are prioritized. This leads to the average waiting time for normal patients to increase drastically, and the system ends up with $W_U = 1/(\mu - \lambda) = 1$ hour and $W_N = \mu/(\mu - \lambda)^2 = 6$ hours. The effect is shown in Figure 2.

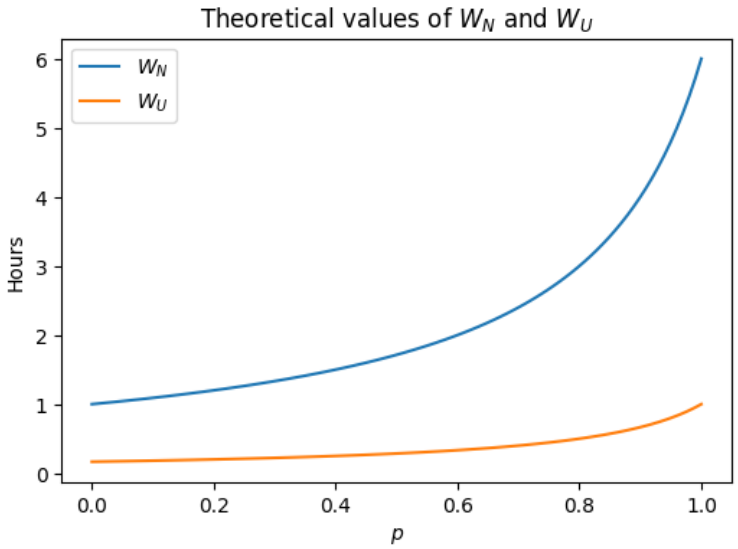


Figure 2: How the expected time in the UCC varies as a function of the proportion of urgent patients

We may calculate the proportion p of urgent patients for which the expected time spent at the UCC for a normal patient is a certain amount. Solving (6) for p:

$$p = \frac{\mu}{\lambda}\left(1 - \frac{1}{W_N(\mu - \lambda)}\right)$$

For $W_N = 2$ hours, the corresponding $p = \frac{3}{5}$.

2 Problem 2: Calibrating climate models

A team of climate scientists is running a deterministic climate model that simulates global temperatures every 6 hours for the years 2006 and 2100. Given fixed initial conditions, external forcing, and model parameters, the output is always the same. However, selecting parameters that yield realistic temperature evolution is challenging, as each model run takes about a month. We assume that the only way to tune parameters is by comparing model output to observed temperatures.

The focus is on a single parameter: the albedo of sea ice, denoted by θ, which measures how much sunlight sea ice reflects. The goal is to choose θ so that model temperatures closely match observations. This match is quantified by a score $\mathrm{y}(\theta)$.

After a month of computation across five centers, the scientists provide five evaluations:

$$(\theta, y(\theta)) : (0.30, 0.50),\ (0.35, 0.32),\ (0.39, 0.40),\ (0.41, 0.35),\ (0.45, 0.60).$$

GP prediction

We model the unknown relationship function between the parameter and its score by a Gaussian process $\{Y(\theta) : \theta \in [0, 1]\}$. We use:

- $\mathrm{E}[Y(\theta)] = \mu \equiv 0.5$,
- $\mathrm{Var}[Y(\theta)] = \sigma^2 \equiv 0.5^2$, and
- $\mathrm{Corr}[Y(\theta_1), Y(\theta_2)] = (1 + 15|\theta_1 - \theta_2|)\exp(-15|\theta_1 - \theta_2|)$.

To process the prediction, we treat $Y(\theta)$ as a multivariate Gaussian on a discrete grid of values of θ, namely from $\theta = 0.25$ to $\theta = 0.50$ with spacing of 0.005. The prior mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ are thus given by:

$$\mu_i = 0.5, \qquad \Sigma_{ij} = \sigma^2 \mathrm{Corr}[Y(\theta_i), Y(\theta_j)]$$

We can now find the posterior mean vector and covariance matrix through transformation of the prior quantities with our observed data.

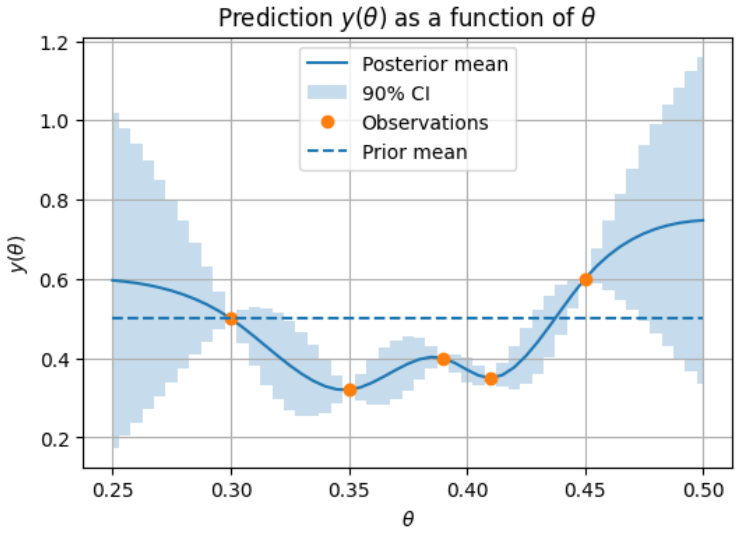


Figure 3: Posterior predictive mean and 90% point-wise intervals for $Y(\theta)$ on the grid $\theta \in [0.25, 0.50]$. Observations are shown with orange dots.

2.1 Probability of achieving the target $y(\theta) < 0.30$

Using the Gaussian process posterior obtained previously, the predictive distribution at any fixed θ is Gaussian:

$$Y(\theta) \mid \text{data} \sim \mathcal{N}\big(\mu(\theta),\, v(\theta)\big),$$

where $\mu(\theta)$ and $v(\theta)$ are the posterior mean and variance at θ. Thus the point-wise conditional probability that $Y(\theta) < 0.30$ equals

$$P\big(Y(\theta) < 0.30 \mid \text{data}\big) = \Phi\left(\frac{0.30 - \mu(\theta)}{\sqrt{v(\theta)}}\right),$$

where Φ is the standard Gaussian cumulative distribution function. The result of evaluating at every θ in our grid is the graph in figure 4.

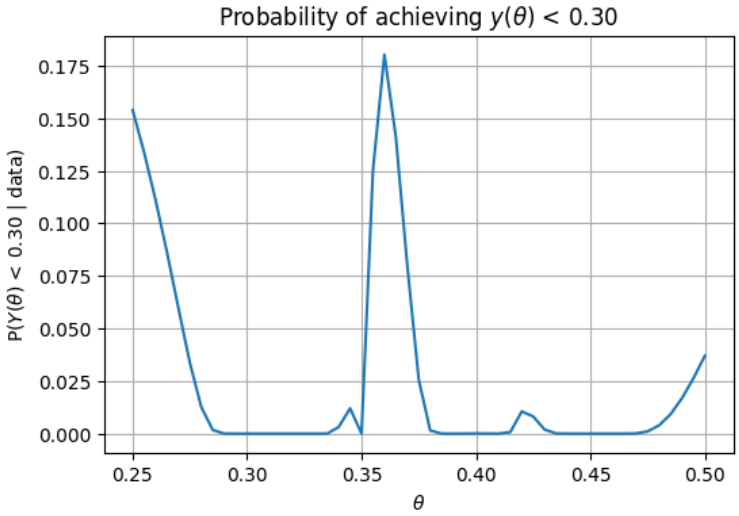


Figure 4: Point-wise posterior probability that $y(\theta) < 30$ as a function of θ.

2.2 New run at $\theta = 0.33$ and choice of the next θ

The scientists perform an additional model run at $\theta = 0.33$ and observe $y(0.33) = 0.40$. We append this new observation to the data set and recompute the GP posterior using the same method.

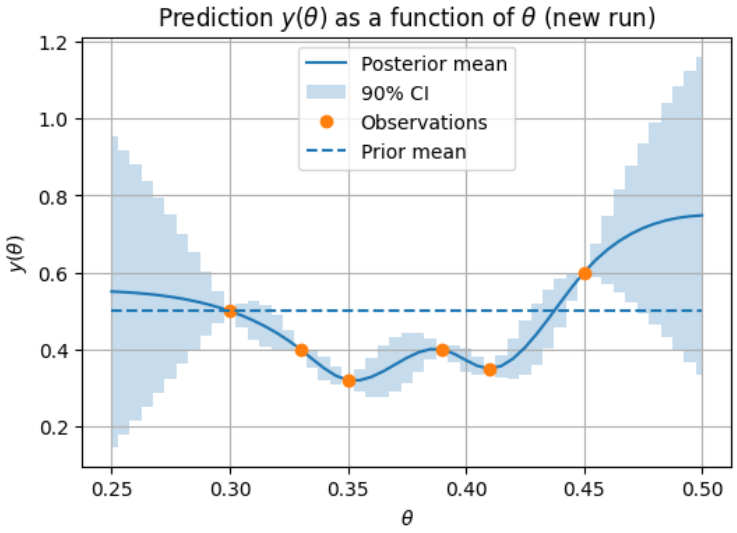


Figure 5: Point-wise posterior probability that $y(\theta) < 30$ as a function of θ, after the new observation at $\theta = 0.33$.

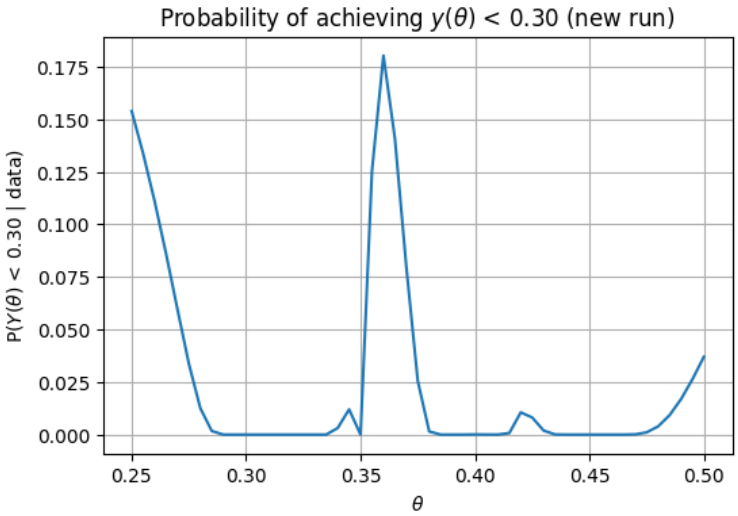


Figure 6: Point-wise posterior probability that $y(\theta) < 30$ as a function of θ, after the new observation at $\theta = 0.33$.

After updating the posterior we again compute the point-wise probabilities $P(Y(\theta) < 0.30 \mid \text{data})$ on the grid. A simple and natural decision rule to choose the next parameter θ to evaluate is to pick the grid point that maximizes this probability (the probability of improvement), which happens to be at $\theta = 0.36$. Intuitively, this selects the input most likely to achieve a $y(\theta) < 0.3$ on the next run.